(3)(a)

Considering $x_1, x_2, \ldots x_n$ are i.i.d distributed.

$$f(x_1 x_2 \ldots x_n / w) = f(x_1/w) \cdot f(x_2/w) \ldots f(x_n/w)$$

$$= \prod_{i=1}^{n} f(x_i / w)$$

now considering a gaussian distribution, likelihood function becomes -

$$p(t_n | x_n, w) = \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp\left(-\frac{(t_n - w^T\phi(x_n))^2}{2\sigma_n^2}\right)$$

Here t_n = observed output for the nth data point.

$w^T\phi(x_n)$ = model's prediction for the nth data point, where $\phi(x_n)$ is a basis function applied to the input x_n, and w is the parameter vector.

σ_n^2 = the variance associated with the nth data point.

[since variance is different for each point]

→ Assuming a Gaussian prior, for a weight vector w, prior is,

$$p(w) = \frac{1}{\sqrt{(2\pi)^N |\Sigma_w|}} \exp\left(-\frac{1}{2}(w-\mu_w)^T \Sigma_w^{-1} (w-\mu_w)\right)$$

⇒ μ_w as the mean of the distribution

⇒ Σ_w as the covariance matrix, that represents the variance and correlation structure b/w the elements of w.

⇒ Σ_w^{-1} is the inverse covariance matrix

⇒ $|\Sigma_w|$ is the determinant of the covariance matrix.